An Automatic Emergency Braking System in CARLA Using Radar–Camera Fusion and Staged PID Control

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***Abstract*—Automatic emergency braking (AEB) must identify a collision-relevant lead vehicle and generate braking early enough to prevent a crash without reacting to adjacent traffic. This paper presents an AEB pipeline implemented in CARLA 0.9.11 for highway car-to-car scenarios. A forward radar is processed at object level through filtering, clustering, and multi-frame tracking. A fine-tuned YOLO26n detector confirms vehicle class in a forward RGB camera. Radar objects are projected into the image and accepted by a geometric fusion gate only when they fall inside a detected car bounding box. A predicted ego-path corridor, time-to-collision (TTC), stopping-distance margin, and a staged PID controller are then used to select a target and command braking. In a final suite of 66 controlled simulations, the system passes all 38 cases within its intended operational design domain (ODD), mainly 50–80 km/h in ideal weather, and passes 63 cases overall. The three retained failures characterize high-speed, short-gap braking and cut-in limits rather than being excluded from evaluation. The results demonstrate a reproducible simulation baseline and explicitly state its limitations; they do not constitute an NCAP certification or real-vehicle validation.**

***Index Terms*—automatic emergency braking, radar–camera verification, target selection, CARLA, vehicle control**

I. Introduction

Forward collision avoidance is a central function of advanced driver-assistance systems (ADAS). Unlike a warning-only system, AEB must transform uncertain perception measurements into a timely braking command. This is challenging in highway traffic because a radar can observe vehicles in adjacent lanes or static roadside structures, while a camera alone does not directly provide reliable relative velocity. Simulation offers a safe, repeatable environment for evaluating these interactions before road testing [1], [2].

This work develops an AEB pipeline in CARLA 0.9.11 using one RGB camera and one forward radar mounted on a Tesla Model 3. The work focuses on car-to-car highway scenarios inspired by the stationary, moving, and braking lead-vehicle groups of Euro NCAP protocols [3]. Cut-in, adjacent-lane, curve, and multi-actor cases are added as engineering stress tests, not as official NCAP tests.

The contributions are: 1) an online object-level radar pipeline with trajectory-corridor gating and multi-frame target confirmation; 2) a camera–radar geometric gate that uses online sensor data and YOLO detections, rather than CARLA actor IDs or ground truth, for AEB decisions; and 3) an end-to-end AEB pipeline that integrates target selection, the fusion gate, TTC, stopping distance, and staged PID braking, evaluated across ODD-oriented and limit-finding scenario sets.

II. Related Work

CARLA provides configurable actors, sensors, synchronous simulation, and ground-truth labels, making it widely used for autonomous-driving research [1], [4]. Radar and camera are complementary: radar measures range and radial relative velocity robustly, whereas visual detectors provide semantic classification. Modern one-stage YOLO detectors make the latter feasible at runtime [5]. Camera–radar fusion commonly relies on calibrated geometric association before tracking or decision fusion [6].

For collision risk, TTC is informative only when the target is closing; stopping-distance criteria additionally account for ego speed and assumed deceleration [7]. This paper uses these conventional principles in a reproducible, explicitly scoped CARLA system. Its contribution is the integration and evaluation of object-level radar processing, image-space confirmation, path-aware target selection, and staged PID braking, not a probabilistic fusion estimator.

III. System Design

A. System Overview

Fig. 1 summarizes the online pipeline. The radar output is filtered by forward range, height, and distance to a predicted ego trajectory. Neighboring points are clustered using spatial, vertical, and relative-velocity thresholds. Each cluster yields a robust measurement: longitudinal position uses the 20th percentile, while lateral position, height, and relative velocity use medians. Tracks are associated frame-to-frame by predicted position and velocity and must persist for multiple frames before they can be selected.

The camera stream is processed by a YOLO26n model fine-tuned for the single *car* class. The final dataset contains 1,505/300/200 train/validation/ test images with 1,872/379/264 boxes. Training at 640×640 for 100 epochs produced precision 0.9919, recall 0.9642, mAP_{50} 0.9940, and $\mathrm{mAP}_{50:95}$ 0.9495 at the best epoch. Ground truth is used only offline to generate labels; it is not available to the online AEB decision.

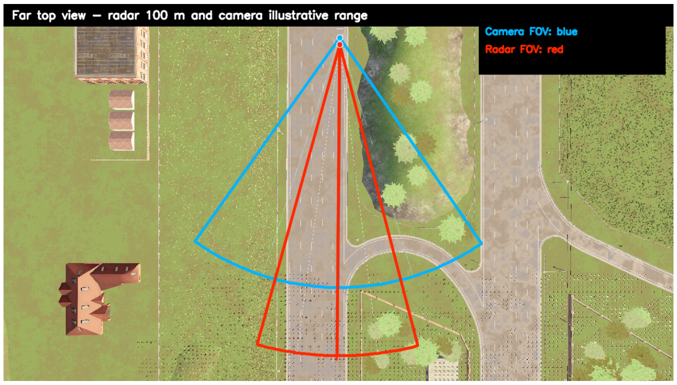

Fig. 1. Top-view sensor coverage in the CARLA ego vehicle. The camera supplies vehicle confirmation; radar supplies range and relative velocity.

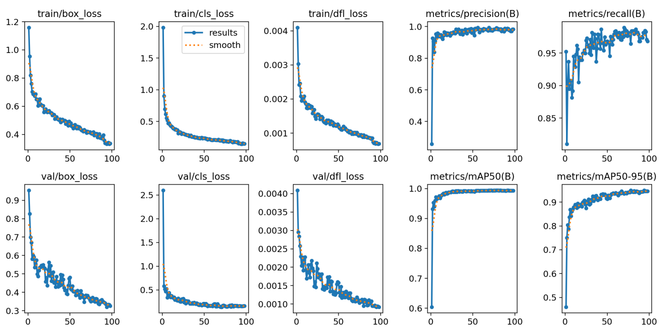

Fig. 2. Training curves of the YOLO26n model used for camera verification.

Fig. 2 summarizes the final training run. The model is used conservatively: it supplies an image-space class check for a target already selected by radar, rather than a stand-alone collision decision. This division keeps the velocity-sensitive risk calculation tied to radar quantities while still rejecting a radar target that is not visually associated with a vehicle.

B. Object-Level Radar Processing

A single radar return is not allowed to trigger braking. Returns first pass a region-of-interest test: they must lie ahead of the ego vehicle, within the configured range and height limits, and near the predicted ego path. Remaining returns are connected into clusters when their planar separation, vertical separation, and relative-velocity difference are sufficiently small. The representative longitudinal coordinate is the 20th percentile of a cluster, which avoids letting returns from the rear of a vehicle push the estimated obstacle farther away. Lateral position, height, and relative velocity use medians instead.

The cluster tracker predicts longitudinal position with relative velocity and associates the next measurement by position and velocity. A track becomes confirmed after three consecutive hits; lost tracks are released after the configured miss limit. This produces a small object list with distance, lateral offset, relative velocity, point support, and track age. The design is not a radar signal-processing model or a probabilistic tracker; it is an explicit object layer placed between CARLA radar returns and the AEB decision.

This object abstraction also makes the decision traceable. The target selector considers only confirmed, non-stale tracks. It first prefers the smallest finite TTC and uses longitudinal distance to break a tie. Hence a lateral reflection or a one-frame point cannot directly override vehicle control. The track confidence used in the user interface is only a deterministic proxy based on point support, recent hits, and freshness; it is not radar cross section, signal-to-noise ratio, or a calibrated existence probability. This distinction matters because CARLA does not expose the raw FMCW signal-processing chain represented by production automotive radar.

C. Path-Aware Target Selection

The nearest object is not necessarily collision-relevant. A constant-curvature path is estimated from ego speed, yaw rate, and steering input. Its horizon is the smaller of the 100-m radar range and a speed-dependent look-ahead distance. Each confirmed object is compared with this path, and objects outside the lateral corridor are rejected before risk evaluation. The remaining candidates are ordered by finite TTC and then longitudinal distance. A selected target must also remain stable for five frames, except when a short-distance or strongly negative stopping-margin condition requires immediate braking. This additional gate separates a persistent lead vehicle from a transient return.

The path representation estimates relevance, not a collision-free trajectory or a lane map. When curvature is close to zero, the path is a straight line; otherwise a point at arc length s follows

$$x(s) = \frac{\sin(\kappa s)}{\kappa}, \qquad y(s) = \frac{1-\cos(\kappa s)}{\kappa}. \tag{1}$$

The closest distance from a radar object to the sampled curve is compared with the lateral AEB limit. This keeps the method independent of CARLA actor identity and lane labels, while making its intended behavior on curves and adjacent traffic inspectable in the logs.

D. Camera Verification

For a radar point transformed to camera coordinates (X_c, Y_c, Z_c), its pixel coordinates are

$$u = f_x X_c/Z_c + c_x, \qquad v = f_y Y_c/Z_c + c_y, \quad Z_c > 0, \tag{2}$$

where the focal length follows $f = W/[2\tan(\mathrm{FOV}/2)]$. A selected radar object is camera-confirmed if its projected point lies inside a non-maximum-suppressed YOLO car box. A 0.35-s confirmation hold absorbs camera–radar timing differences. Radar remains the source of range and relative velocity, while the camera prevents an unconfirmed radar target from initiating braking. In particular, the camera does not estimate range or relative velocity, and this late geometric gate is not presented as joint radar–camera tracking.

At each tick, a radar decision outside the supervisory 'BRAKE' state passes unchanged. If radar requests 'BRAKE', the gate permits it only when the selected target is currently inside a valid vehicle box or was confirmed within the 0.35-s hold window. Otherwise the decision is converted to 'RELEASE' with zero brake. The hold avoids a rapidly toggling command when camera, radar, and YOLO arrive on different frames. It also exposes an important trade-off: a missed camera detection can delay braking, which is treated as a limitation rather than hidden by a radar-only fallback in the evaluated configuration.

E. Risk Assessment and Staged PID Braking

The predicted path is a constant-curvature curve estimated from ego yaw rate and steering. Objects outside its lateral corridor are rejected. For a target at distance d with relative velocity v_r, closing speed and TTC are

$$v_c = -v_r, \qquad \mathrm{TTC} = d/v_c \quad (v_c > 0). \tag{3}$$

TTC is infinite for a non-closing target. The required stopping distance is

$$d_{\mathrm{req}} = \max\left(d_0, v_e t_r + \frac{v_e^2}{2a_e} - \frac{v_t^2}{2a_t} + d_0\right), \tag{4}$$

where v_e, v_t are ego and target speeds, t_r is response time, a_e, a_t are assumed emergency decelerations, and d_0 is a safety offset.

The controller uses both time-based and distance-based risk because neither is sufficient alone. TTC is not a hazard measure for a target moving away, whereas stopping distance accounts for ego speed, assumed response delay, and the target's own ability to decelerate. The distance margin is

$$m_d = d - d_{\mathrm{req}}. \tag{5}$$

An active 'BRAKE' state is therefore requested either by a low TTC or by an insufficient margin. The PID input combines a positive margin deficit with a positive TTC deficit. Its integral is clamped, the derivative is used only when error is increasing, and the output is rate-limited. These choices avoid a large release–reapply oscillation in discrete 20-Hz control; they do not claim an optimal vehicle-braking controller.

The supervisory state machine contains 'NORMAL', 'WARNING', 'BRAKE', and 'RELEASE'. Soft, medium, hard, and emergency are brake-command schedules inside 'BRAKE', rather than four independent states. The controller enters 'BRAKE' from TTC or stopping-margin risk. Within it, the PID command is rate-limited and clipped to the applicable brake interval. This retains a continuous command while allowing immediate maximum braking in an emergency.

IV. Experimental Setup

The simulator runs synchronously at 20 Hz in Town04 under ideal weather. The ego vehicle is a Tesla Model 3 with a windshield RGB camera and a bumper-mounted forward radar. A hazardous scenario passes when AEB brakes without collision; a non-hazardous scenario passes only if it produces no false brake. Logged metrics include collision status, minimum gap, first-brake distance, ego speed, TTC, and brake command.

TABLE I
KEY SIMULATOR AND CONTROLLER PARAMETERS.

Parameter	Value
RGB camera / horizontal FOV / tick	1280×720 / 70° / 0.05 s
Radar range / horizontal–vertical FOV	100 m / 30°–6°
Radar points per second / tick	2000 / 0.05 s
Track confirmation / target confirmation	3 / 5 frames
Warning / brake TTC	3.0 / 1.5 s
Response time / ego deceleration	0.20 s / 8.0 m/s^2
Staged brake: soft / medium / hard / emergency	0.55 / 0.75 / 0.90 / 1.00

TABLE II
EVALUATION SCOPE AND INTERPRETATION.

Set	Typical conditions	Purpose
Designed ODD	Highway, cars only, ideal weather, mainly 50–80 km/h	Verify target operating range
Stress tests	95–110 km/h, short gaps, hard braking, difficult cut-in	Identify operating limits

The final evidence suite contains CCR stationary (CCRs), moving (CCRm), and braking (CCRb) cases plus clear-road, adjacent-lane, curve, cut-in, cut-out, and multi-actor cases. Table II separates the intended ODD from stress tests. This avoids interpreting a single aggregate pass rate as an unconditional safety claim.

The final suite uses the YOLO ONNX model, the radar-object pipeline, geometric camera verification, and the staged-PID configuration. A hazardous case passes only when no collision is reported; a non-hazardous case passes only when the system does not brake unnecessarily. The fixed-tick logs, rather than display frame rate or video playback, provide the quantitative evidence. The suite is deterministic in the simulator and is used to map behavior in configured cases, not to estimate statistical reliability.

The scenario groups have separate purposes. CCRs tests a stationary lead vehicle that stays relevant throughout the run. CCRm tests a slower lead vehicle, so the closing speed can be much smaller than ego speed. CCRb introduces a lead vehicle that brakes after both vehicles are moving. Cut-in and cut-out test changes in path relevance, while adjacent-lane, curve, and multi-actor cases exercise the ability to ignore a nearby but non-relevant object. This family-based setup is inspired by car-to-car evaluation structures, but it does not reproduce the full test apparatus, measurements, or certification conditions of Euro NCAP.

V. Results and Discussion

A. Overall Outcomes

Table III reports the final staged-PID, camera–radar system. All 38 cases in the designed ODD pass. The full suite contains 63 passes among 66 cases. The three collisions are retained:

TABLE III
FINAL RESULT WITH CAMERA–RADAR FUSION AND STAGED PID.

Evaluation set	Cases	Pass	Fail
Designed ODD	38	38	0
Stress / limit finding	28	25	3
Total	66	63	3

TABLE IV
CCRB SWEEP: P = PASS AND C = COLLISION.

Ego speed / gap	20 m	30 m	40 m	50 m	60 m	80 m
50 km/h	P	P	P	P	P	P
65 km/h	P	P	P	P	P	P
80 km/h	P	P	P	P	P	P
95 km/h	C	P	P	P	P	P
110 km/h	C	P	P	P	P	P

CCRb at 95 and 110 km/h with a 20-m initial gap, and a 100/60-km/h cut-in with a 25-m initial gap. They reveal that, once the lead vehicle brakes or enters the ego corridor late, the available perception and stopping time becomes insufficient.

B. Behavior Across Scenario Families

The stationary, moving, braking, and cut-in families reveal different forms of risk. For CCRm, absolute ego speed alone is not decisive: the 110/80-km/h, 20-m case passes because its closing speed is about 30 km/h. By contrast, a stationary lead vehicle or a braking lead vehicle can remove closing-speed margin quickly. The configured clear-road, adjacent-lane, curved-road, and multi-actor cases provide complementary checks: no unnecessary braking was observed for the specified adjacent-lane and curved-road cases, while same-lane hazards still triggered braking. These are configured-case observations, not an ablation that attributes a quantitative reduction in false brakes to a single component.

The representative values illustrate the same point. In the 80/50-km/h CCRm case, braking begins at a gap of 18.97 m and the smallest measured gap is 14.45 m. In the 110/80-km/h CCRm case, braking starts at 18.96 m and the minimum gap is 15.01 m. Both are different from the CCRb cases even though some initial gaps are identical, because the leading vehicle continues moving instead of removing its speed. The result motivates retaining relative velocity and stopping margin in the controller rather than using a fixed distance threshold.

C. Observed Failure Boundaries

Fig. 3 contrasts a successful 80-km/h CCRb case with its 95-km/h limit case. At 80 km/h and a 20-m initial gap, braking begins at 18.56 m and the minimum gap remains 1.96 m. At 95 km/h under the same gap, the controller still commands braking but collision occurs with a 0.013-m minimum gap. Likewise, the 100/60-km/h cut-in is detected only after the entering vehicle reaches the ego corridor, so braking begins at 7.26 m and is too late to avoid contact.

The CCRb sweep in Table IV makes the stopping-distance boundary visible: the 20-m initial gap passes at 80 km/h but

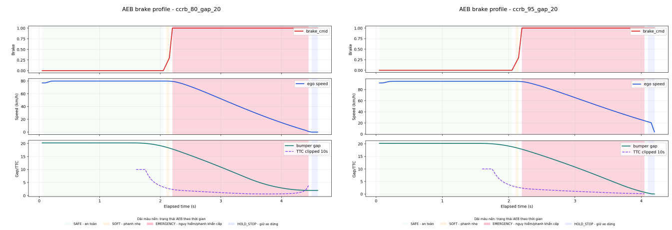

Fig. 3. Brake profiles for CCRb at a 20-m initial gap: pass at 80 km/h (left) and collision at 95 km/h (right).

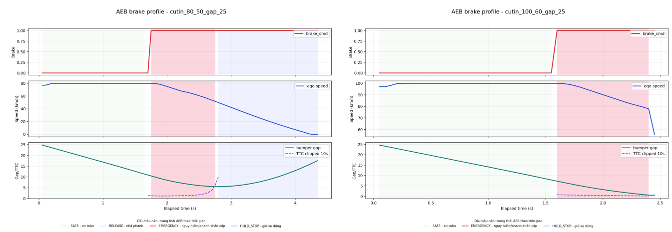

Fig. 4. Cut-in brake profiles at a 25-m initial gap: pass at 80/50 km/h (left) and collision at 100/60 km/h (right).

collides at 95 and 110 km/h, whereas the reported larger gaps pass. The evidence does not show that perception failed in those cases; the final controller still commands braking. It supports a more limited conclusion: in this configured braking scenario, the available distance becomes insufficient as speed increases. The cut-in failure has a different mechanism. The target begins outside the ego corridor, so it becomes relevant only after entering that corridor. At 100/60 km/h with a 25-m initial gap, confirmation and braking begin at 7.26 m, leaving too little distance. Table V places these observations alongside two successful cases. Its values reinforce that the result should not be read as a single speed threshold: distance, target motion, and when a vehicle becomes path-relevant all affect the remaining stopping margin.

The late-fusion gate has a deliberate asymmetry. Radar can establish that an object is closing and within the predicted path, but the gate still requires a recent image-space vehicle match before allowing a brake command. This reduces the chance that an unclassified return alone becomes a braking target in the configured adjacent-lane cases. The corresponding cost is visible in cut-in: when a vehicle crosses into the ego corridor late, the first relevant radar object and the first usable camera confirmation are both delayed. The 80/50-km/h case retains enough distance after braking; the 100/60-km/h case does not.

This observation should not be generalized into a claim that camera verification is always beneficial or that the configured path corridor is optimal. The final evidence contains no repeated radar-only versus camera-gated comparison. It instead documents a transparent behavior: a selected radar target must remain kinematically relevant and visually confirmed, and the available margin can be exhausted if that relevance appears late. A future estimator could combine lateral motion, predicted lane entry, and uncertainty-aware association, but

TABLE V
REPRESENTATIVE FINAL-EVIDENCE SCENARIOS. BRAKE QUANTITIES ARE SAMPLED AT THE FIRST NONZERO BRAKE COMMAND.

Family	Scenario	Result	Brake speed (km/h)	Brake gap (m)	Minimum gap (m)
CCRs	ccrs_80	Pass	69.72	32.12	7.70
CCRm	ccrm_110_80_gap_20	Pass	108.00	18.96	15.01
CCRb	ccrb_80_gap_20	Pass	79.92	18.56	1.96
CCRb	ccrb_95_gap_20	Collision	94.90	18.56	0.013
Cut-in	cutin_80_50_gap_25	Pass	79.92	10.46	5.51
Cut-in	cutin_100_60_gap_25	Collision	99.90	7.26	0.476

would need its own controlled evaluation.

The controller outcome is also best interpreted through the stopping model. The configured ego deceleration is 8.0 m/s^2 and the response term is 0.20 s, but these are risk-model parameters inside CARLA rather than measurements of a brake system. When a target is moving, the model subtracts its estimated stopping distance; this is why a moderate-closing-speed CCRm case can retain a large minimum gap even at high ego speed. When a CCRb target begins braking, its future speed changes faster than a constant-relative-velocity TTC picture. The margin and staged emergency limits provide an additional trigger, but neither can create physical distance after the target becomes relevant. The boundary cases show where the configured chain of perception, decision, and actuation runs out of time.

The evidence also separates visualization from measurement. The graphical user interface displays radar points, selected objects, YOLO boxes, target status, TTC, and brake command, but the evaluation relies on synchronous simulation ticks and generated logs. Video frame rate can differ from the 20-Hz simulator tick and is not used as evidence for response time. Similarly, raw jerk derived from CARLA state is retained for internal controller comparison but is not interpreted as an on-road ride-comfort value. This separation keeps the paper focused on outcomes that the project records consistently across the final suite.

VI. LIMITATIONS

The findings are limited to CARLA 0.9.11, Town04, ideal weather, and car-to-car scenarios. CARLA radar supplies simulated detections rather than a raw FMCW signal chain, and the YOLO dataset is concentrated in the same simulation domain. The camera gate can withhold a brake if a detection is missed or calibration is misaligned. No randomized repeated trials or quantitative radar-only/fusion ablations are available in the final evidence, so the study does not estimate variance or compare alternatives statistically. Pedestrians, motorcycles, adverse weather, actuator delay, tire-road interaction, and full ABS dynamics are outside the evaluated scope. These limitations identify the next validation steps; they do not turn the simulated result into a vehicle-safety claim.

The most direct extensions are repeated trials with controlled seeds, explicit radar-only and camera-gated baselines, and reporting of collision, minimum gap, and unnecessary-brake distributions. A second group of extensions concerns the late cut-in boundary: lateral-motion prediction and time-to-lane-entry estimates could make an entering vehicle relevant before it occupies the center of the ego corridor. Neither follows automatically from widening the corridor, because that would also increase exposure to adjacent-lane false targets. A more realistic evaluation would also need diverse maps, lighting and weather, pedestrians and two-wheelers, actuator delay, and tire-road dynamics.

The detector result is also domain-limited. YOLO26n was fine-tuned with CARLA-derived labels for one 'car' class, mainly in Town04 and favorable visual conditions. Its validation precision, recall, and mAP support that narrow verification role, but they do not establish transfer to another map, night, rain, pedestrian targets, or camera hardware. The radar-side filters are equally configuration-dependent: their range, height, and lateral thresholds were chosen for the present vehicle and highway scenarios. Publishing these parameters and the exact scenario split is consequently more useful than presenting the result as a general AEB performance number.

VII. CONCLUSION

This paper presented a CARLA AEB system that fuses object-level radar processing with YOLO vehicle confirmation, path-aware target selection, TTC and stopping distance, and staged PID braking. It passes 38/38 cases in the stated ODD and 63/66 cases in the complete limit-finding suite. The retained failures define the current high-speed, short-gap, and late cut-in limitations. Future work will add repeated-seed ablation studies, diverse weather and maps, probabilistic multi-object fusion, and vehicle/actuator dynamics before any real-world claim.

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